

- as per your requirements.
- 3. The only security parameters you set are for your own app. The cloud security is dealt with by your cloud service provider.
- 4. It is time- and cost-effective for companies and individuals looking to host their apps in the cloud, especially startups that cannot afford to build their own infrastructure.

Infrastructure as a Service (IaaS)

Infrastructure as a Service goes one step deeper than PaaS, providing customers with even more autonomy. In an IaaS model, the cloud service provider gives you control over the underlying infrastructure of the cloud. Simply put, you get to design your own cloud environment customised to your company's requirements all the way from dedicated servers and virtual machines, to operating systems running on the servers, setting bandwidths, creating your own security protocols, and pretty much everything else that goes into creating a cloud infrastructure. Amazon AWS and Google Compute Engine are great examples of IaaS models. Given the autonomy over the hardware that this model provides, it is also referred to as Hardware as a Service (HaaS).

Benefits of IaaS

1. Granular flexibility in the 'pay as you go' model. You get to decide how many VMs you want to run and for how long. You can even pay by the hour.
2. Highly scalable, given that it follows the 'pay as you go' model to its core.
3. Complete autonomy and control over everything in the infrastructure without the hassle of maintaining the servers physically at your company location.
4. Most companies guarantee uptime, security and 24/7 on-site customer

support, which can be essential for enterprises.

Storage as a Service (StaaS)

Google Drive, OneDrive, Dropbox, and iCloud are some of the big names in the industry providing Storage as a Service to their customers. StaaS is as simple as it sounds. If all you require is storage in the cloud that is accessible to you in real-time through any of your devices, then the StaaS model is the one to choose. Many companies and individuals make use of this service model to back up their data.

Benefits of StaaS

1. Access your data in its most updated form in real-time with the help of built-in version control systems.
2. Access your data through any type of device with any operating system.
3. Back-up your data in real-time as and how you create, edit, and delete your files.
4. Scale your storage as and how you require. StaaS follows the 'pay as you go' model.

Anything/Everything as a Service (XaaS)

A hybrid version of the IaaS, PaaS, SaaS, and StaaS, is what is being called the Anything/Everything as a Service model, and is quickly gaining traction in the cloud community. It is possible for a customer to have requirements that are so varied that they might be a mishmash of all the different models. In such a scenario, complete autonomy is provided to customers to select the services from different tiers to create their own custom 'pay as you go' model. This has the benefit of giving complete freedom to the customer to use the cloud on their own terms.

Benefits of XaaS

1. Choose what you like, how you like and as you like.

2. Pay only for exactly what you need without having to pay for any base fee predicated on a tier system.
3. Select your infrastructure, platform, and functionality on a granular level.
4. If used appropriately, XaaS can be the most time-, cost- and work-effective method of hosting your application on the cloud.

Function as a Service (FaaS)

In certain cases, companies or individuals require the benefits of PaaS without having to use all its functionality. For example, trigger-based systems such as cron jobs only require a piece of code or a function to run on a serverless system to achieve a particular objective. For instance, a customer may want to create a website traffic monitoring system that sends a notification the moment a certain number of page downloads occur. In such a case, the customer requirements are simply to run a piece of code in the cloud that keeps checking for a trigger to execute. Using a PaaS model can be a costly solution. This is where Function as a Service comes in. Many companies such as Heroku offer FaaS to their customers to host only a specific piece of code or function that is reactionary and only activates upon a trigger.

Benefits of FaaS

1. You only pay for the number of executions of the code. You are generally not charged for hosting your code unless it is computationally expensive.
2. It removes all the liability of PaaS while giving you all its benefits.
3. You are not responsible for the underlying infrastructure in any way. Therefore, you can simply upload your code without having to worry about any maintenance of the virtual machines.

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